

UAVs

Aeronaves Robotizadas

MEAer, 1st year, 1st sem 2022/23
conclusion

José Raul Azinheira

January 2023

UAVs

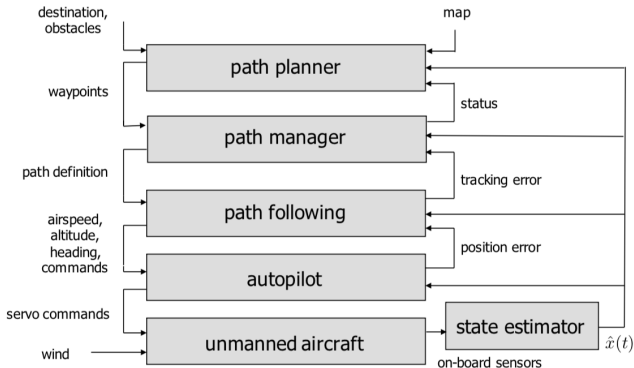
Simulation and control

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Conclusion ?

overall Beard block diagram



Small Unmanned Aircraft: Theory and Practice

Randal W. Beard & Timothy W. McLain, 2012 Princeton University Press

UAVs

Simulation and control

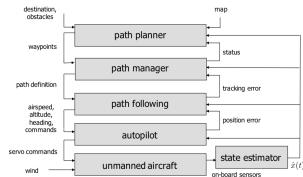
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Conclusion ?

we have been
talking of: **modeling the system**

- modeling the UAV motion
 - its actuation
 - its simulation
- state estimation



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Simulation and control

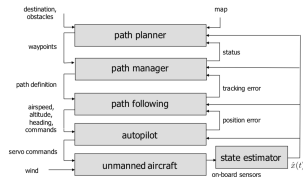
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Conclusion ?

we have been talking of: **control**

- modeling the UAV motion (and then its simulation)
- state estimation
- control techniques: linear and nonlinear
- control architectures: PID, cascade



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Simulation and control

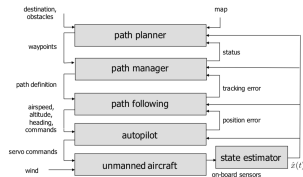
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Conclusion ?

we
 have been talking of: **path following**

- modeling the UAV motion (and then its simulation)
- state estimation
- control techniques: linear and nonlinear
- control architectures: PID, cascade
- path following
- path management / planning



UAVs

Simulation and control

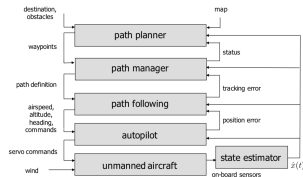
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in the lab sessions: **hands on**

- Matlab/simulink tool
- modeling/simulation
- estimation
- path control



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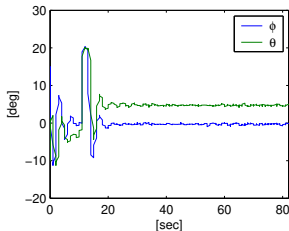
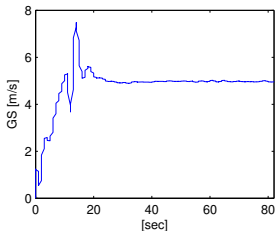
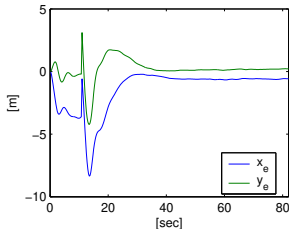
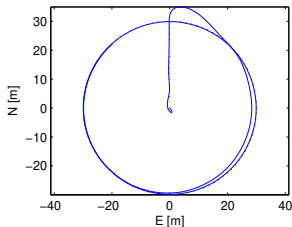
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lab 3: example circle path control similar to : head towards origin



Conclusion ?

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Application needing a UAV or any mobile robot:

- **requirements for the robot: top down logic**
 - payload: application sensors (pan-tilt camera?)
 - navigation and planning needs
 - UAV platform options: e.g. fixed wing or multi-rotor
 - robot sensors and actuators

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- **system design: bottom up logic**

- hardware specs and integration
progress in size and costs have had a dramatical impact
- low level integration
modeling and simulation are a fundamental tool
- software and navigation architecture
modularity is a key characteristic
- motion/application integration
it is not only a problem of successfully moving from A to B

summary of the sequence:

Chapter 0: Introduction

Chapter 1: Rigid body EOM

Chapter 2: Quadrotor modeling

Chapter 3: Quadrotor control introduction

Chapter 4: Modern Control Design

Chapter 5: Sensors

Chapter 6: State estimation

Chapter 7: Nonlinear control

Chapter 8: Path following

Chapter 9: Path management